

1 Milestones and Team Roles

1.1 Project Milestones Review

This project has successfully processed the key milestones defined in the Proposed Design report. More details can be identified as follows:

M1: Data Input standardization (Week 1): Cleared definition of temporal network standards and implementation of formatted data uploading and parsing. Established the (u, v, t) data structure and parsing logic.

M2: Algorithm Verification (Week 4): Successfully processed implementation and verification of the temporal reachability BFS. Set up static data frame for the data processing.

M3: System Integration (Week 6): Developed the integration of UI and the verified temporal reachability algorithm. Achieved visualization by using Streamlit front-end.

M4: Time Control Mechanism (Week 6): Developed the cumulative time slider for frame-by-frame progression.

M5: what-if Functionality (Week 7): Implemented the What-if feature to edit selected edges.

1.2 Extent to which Aims and Objectives were Met

All primary aims and objectives were fully met. The application successfully handles datasets of up to $N = 500$ nodes and $M = 2000$ edges while maintaining stable performance. The What-if tool provides immediate feedback on global reachability metrics, which achieved effectively visualization of the dynamic temporal path network.

1.3 Team Members and Roles

Table 1: Detailed Coding Function Ownership

Team Member	Primary Role	Key Function
Yiqing LIU	System Integration	<code>build_interface</code> (UI Layout & Wiring)
Ankit RANJAN	Data Engineering	<code>standardize_data</code> (Cleaning & Sorting)
Xiaohan XU	Core Algorithms	<code>calculate_earliest_arrival</code> (Temporal BFS Algorithms)
Bowen LI	Simulation Dev	<code>apply_what_if_modifications</code> (Scenarios)
Zixuan TANG	Optimization	<code>find_menger_bottlenecks</code> (Bottleneck Scan)
Ce WANG	Layout Design	<code>compute_static_layout</code> (Coordinate Mapping)
Wenbin ZENG	Visual Interaction	<code>highlight_active_elements</code> (UI Enhancement)

1.4 Individual Role Summaries

- **Yiqing LIU:** Main control of the project design and management. Responsible for final integration and overall code review
- **Ankit RANJAN:** Responsible for data cleaning and standardization. Designed the input api.

- **Xiaohan XU:** Developed the BFS algorithms structure, which implements the earliest arrival search mechanism.
- **Bowen LI:** Developed and tested the What-if functionality.
- **Zixuan TANG:** Code optimization.
- **Ce WANG:** Integrated Streamlit front-end to the project and designed the static layout data frame.
- **Wenbin ZENG:** UI and interface optimization